

304 Quad Channel Voltage Controlled Mixer User Manual

05-2026 Rev. 1

1 Introduction

The *304 Quad Channel Voltage Controlled Mixer* is a flexible mixer module for audio. It comprises of four linear VCAs based on LM13700 dual OTA ICs. Each channel can be individually routed out through channel outputs, and are normalled to be mixed and output at the fourth output. This means that the modules can be used as anything between four individual VCAs and a quad channel mixer.

2 Installation & Power

This module requires 10hp of free space in a rack. When installing, always turn off the power supply before plugging anything in. When plugging in the 10-16 pin ribbon power cable, ensure that the red stripe lines up with the **RED LINE** mark (refer to fig. 2.1) on the module and the marking that indicates **-12v** on the bus board.

This module **is not** reverse polarity protected. This means that the module will get heavily damaged if the power cable is plugged in backwards. Additionally, a reversed power cable may still cause damage to other modules connected to the same power supply.

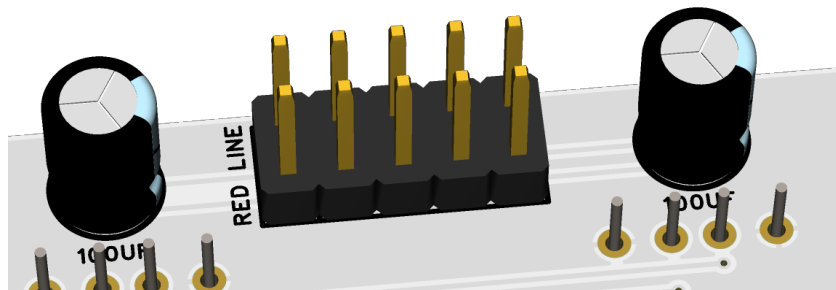


fig. 2.1 Power Connector

3 Front Panel Overview



fig. 3.1 Panel Overview

The 304 panel lacks labels and text, but is still fairly straightforward to use. The elements in each channel is laid out vertically in strips, and four of those identical channels are laid out next to each other.

For each channel, there are 2 inputs and 1 output. The topmost input is the VCA **CV** in, with its respective attenuator right below it. The slider in the middle is the channel **VOLUME** slider.

Note: The **VOLUME** slider is a voltage offset, and will not be able to control the volume when **CV** is applied (see fig. 3.2).

Below the **VOLUME** slider are the audio **IN** and channel **OUT** jacks. If nothing is plugged into the **OUT** jack of a channel, the input will get normally mixed into the channel four **MIX OUT** (the rightmost output).

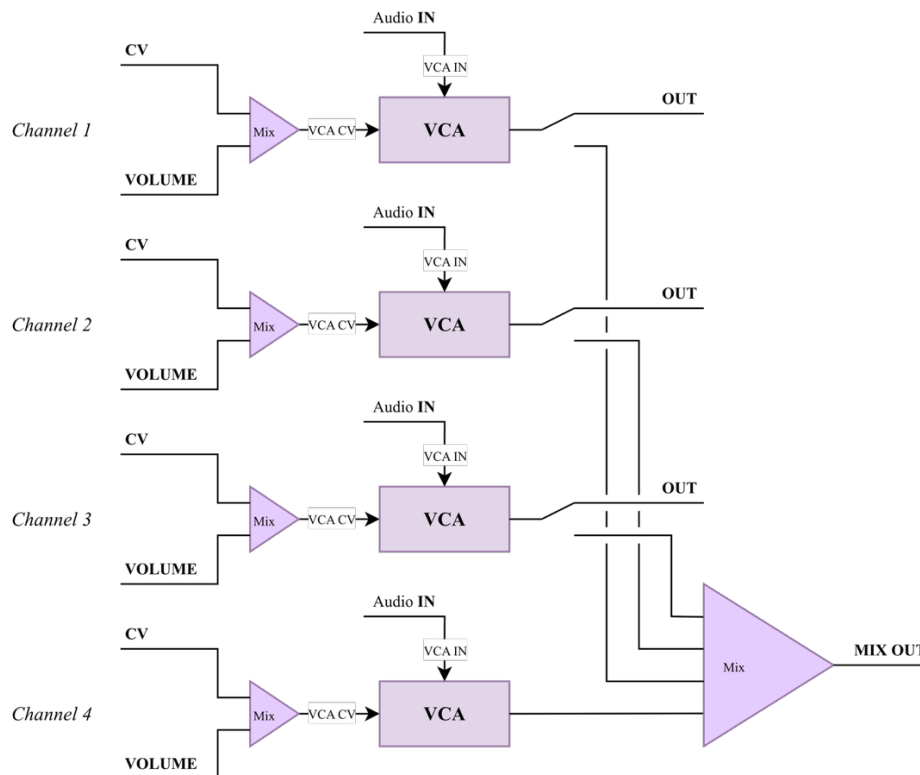


fig. 3.2 Block Diagram

4 Additional Information (Optional)

bleedthru

5 Technical Specifications

5.1 Dimensions & Power

Width	Depth	+12v Current	-12v Current
<i>10hp</i>	<i>~35mm max.</i>		

5.2 Inputs & Outputs

Inputs		Outputs	
IN 1 - 4	<i>Any modular signal levels</i>	OUT 1 - 4	<i>~20 V_{pp} max.</i>
CV 1 - 4			

5.3 Other Specifications